



CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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H2 Biology

9744/02

Paper 2 Structured Questions

14 September 2022

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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Total	100

This document consists of **30** printed pages and **2** blank pages.

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Answer **all** questions.

- 1 The Golgi body, rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) form part of the internal membrane system of a cell. The membranes have a fluid mosaic structure.

Fig. 1.1 is a transmission electron micrograph of one area of a liver cell showing a region with RER and a region with SER. Mitochondria are also visible in the image.

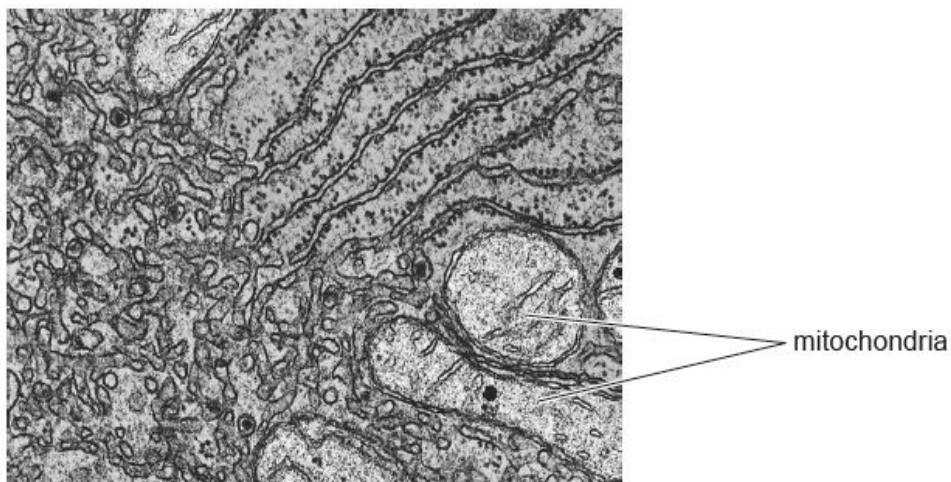


Fig. 1.1

- (a) Describe **one** difference between RER and SER.

structure

.....

function

.....[2]

- (b) Phospholipids are one of the main components of membranes.

Describe the structure of a phospholipid molecule.

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..... [2]

(c) One function of a Golgi body is to package molecules into Golgi vesicles.

(i) A Golgi body and Golgi vesicles are **not** visible in Fig. 1.1.

Describe the features, **other than** the presence of Golgi vesicles, that would help you identify a Golgi body in a transmission electron micrograph of another area of the same liver cell.

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.....[2]

(ii) Some Golgi vesicles contain secretory proteins for release from the cell.

Describe the sequence of events that occurs following the packaging of a secretory protein into a Golgi vesicle to its release from the cell.

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.....[3]

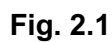
(iii) Some Golgi vesicles contain glycoproteins or glycolipids to be added to the cell surface membrane.

State a role of **glycolipids** in the cell surface membrane.

.....

.....[1]

[Total: 10]



[Turn over]

The feet of elephants are protected by structures under the skin known as cushions, as shown in Fig. 2.2.



Fig. 2.2

The cushions are made up of a large number of cells surrounded by connective tissue containing many fibres of collagen. The fibres help to maintain the structure of the cushion.

The collagen fibres are made of collagen molecules.



Fig. 2.3

With reference to Fig. 2.3,

(b) (i) describe the structure of a wild-type collagen molecule.

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.....[4]

- (ii) Osteogenesis imperfecta (OI), commonly known as "brittle bone disease", is a disorder characterized by bone fragility and abnormalities of connective tissue. Vast majority of affected individuals have mutations in the genes encoding the polypeptide chains of collagen, which result in substitution of amino acid residues.

Suggest the effect of such mutations on the structure of collagen fibre.

.....

[2]

Collagen is a water-insoluble fibrous protein that can be prone to swelling. Collagen swelling occurs when water fills up the gaps between collagen molecules.

Fig. 2.4 shows the effect of pH on swelling ratio of collagen. Swelling ratio is defined as the percentage increase in the weight of collagen due to water absorption.

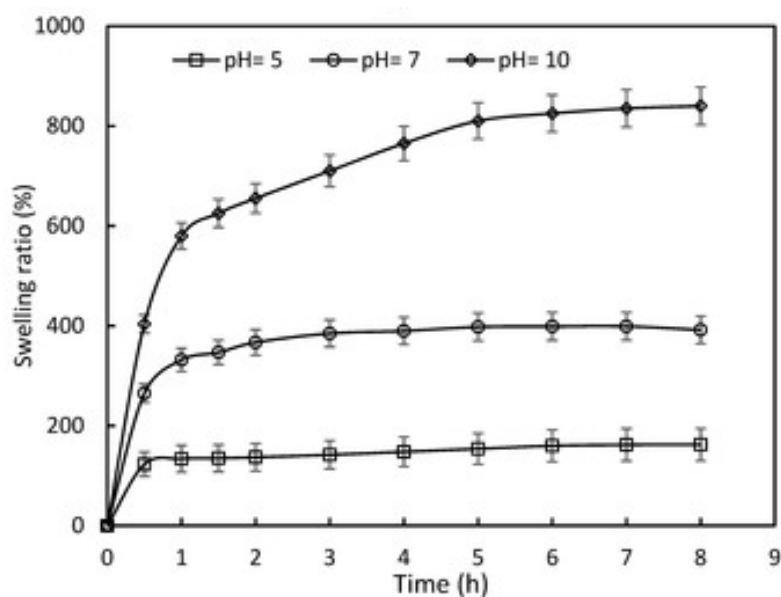


Fig. 2.4

- (c) With reference to Fig. 2.4, describe and explain the effect of pH on collagen.

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 [2]

[Total: 10]

[Turn over]

- 3 (a) Fig. 3.1 is a photomicrograph of root tip cells at different stages in the cell cycle. A cell cycle in interphase is labelled.

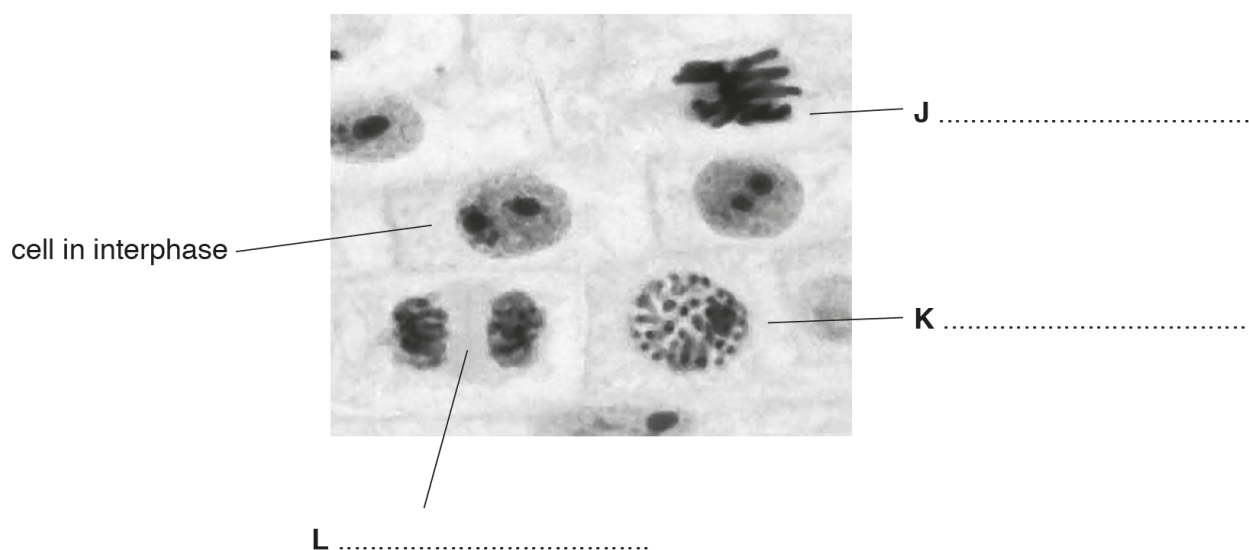


Fig. 3.1

- (i) Name the stage of mitosis shown in each of the cells **J**, **K** and **L** in Fig. 3.1.

Write your answer in the space next to each letter on Fig. 3.1.

[2]

- (ii) Explain how it is possible to deduce that the labelled cell in interphase shown in Fig. 3.1 is in late, rather than early, interphase.

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.....[1]

- (iii) Suggest one advantage of using a light microscope, rather than an electron microscope, to study cell division.

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.....[1]

(b) Mutations in human body cells can sometimes result in a tumour. Some tumours are cancerous.

(i) Outline how tumour development is a multistep process.

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.....[3]

(ii) Tumour cells have antigens on their cell surface that are not present on non-tumour cells.

These antigens are the result of gene mutations and are known as tumour specific antigens (TSA).

One type of TSA differs in structure from the protein found on the cell surface of non-tumour cells by a single amino acid.

Explain how a gene mutation could result in the production of this TSA.

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.....[2]

(c) Immunotherapy is a form of treatment for cancer which aims to stimulate the immune system to destroy tumour cells.

One form of immunotherapy for cancer uses a vaccine which contains one specific type of TSA.

Suggest how vaccination with a specific type of TSA could lead to the destruction of tumour cells by T-lymphocytes in the body.

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..... [1]

[Total: 10]

- 4 Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but all catalyse the same reaction.

Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide: polypeptide M, which is coded for by the LDH-A gene and polypeptide H, which is coded for by the LDH-B gene.

Table 4.1 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

Table 4.1

isoenzyme	polypeptide composition of enzyme	example of isoenzyme location
LDH-1	HHHH	heart red blood cells
LDH-2	HHHM	heart red blood cells
LDH-3	HHMM	brain lungs
LDH-4	HMMM	kidneys placenta
LDH-5	MMMM	liver skeletal muscles

- (a) With reference to Table 4.1, suggest how different cells of the same individual can produce different isoenzymes.

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.....[2]

- (b) The base sequences of the LDH-A and LDH-B genes and the sequences of the amino acids encoded by these genes were determined.

Fig. 4.1 shows the first ten amino acids of polypeptides M and H and the corresponding base sequences of one of the DNA strands of each gene.

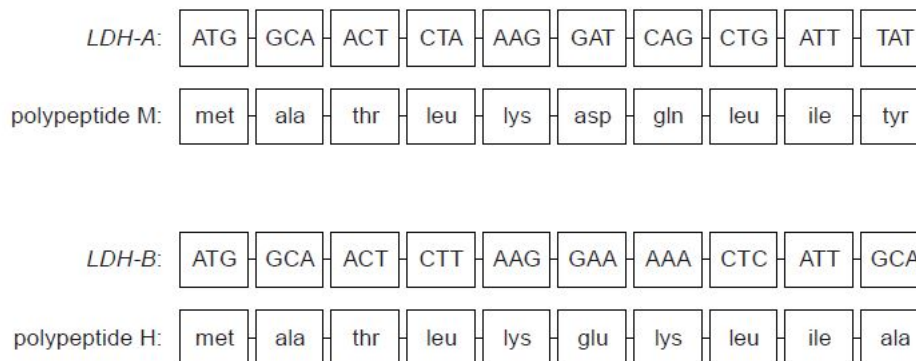


Fig. 4.1

Table 4.2 shows the genetic code (mRNA codons).

Table 4.2

first position	second position				third position
	U	C	A	G	
U	phe	ser	tyr	cys	U
	phe	ser	tyr	cys	C
	leu	ser	STOP	STOP	A
	leu	ser	STOP	trp	G
C	leu	pro	his	arg	U
	leu	pro	his	arg	C
	leu	pro	gln	arg	A
	leu	pro	gln	arg	G
A	ile	thr	asn	ser	U
	ile	thr	asn	ser	C
	ile	thr	lys	arg	A
	met	thr	lys	arg	G
G	val	ala	asp	gly	U
	val	ala	asp	gly	C
	val	ala	glu	gly	A
	val	ala	glu	gly	G

With reference to Fig. 4.1 and Table 4.2,

- (i) state if both the base sequences of LDH-A and LDH-B genes are the template strand or non-template strand of DNA.

..... [1]

(ii) explain your answer to (b)(i).

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..... [2]

(iii) Fig. 4.2 shows the process of polypeptide synthesis.

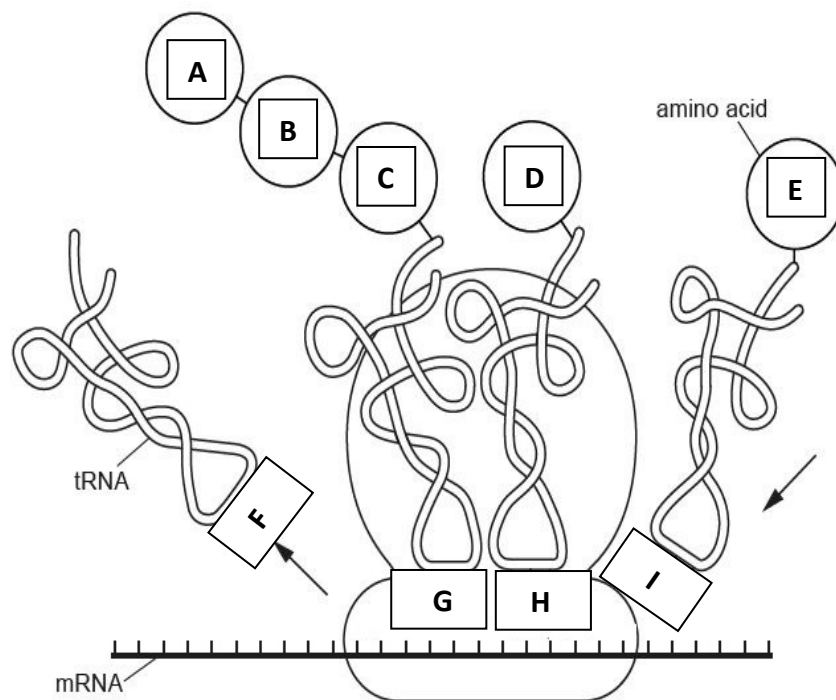


Fig. 4.2

Fill in the following spaces with the relevant information for the synthesis of the first five amino acids for the LDH-A polypeptide chain.

Amino acids: A B C D E

[1]

Anticodons: F G H I

[1]

(iv) Compare the structures of mRNA and tRNA.

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..... [3]

[Total: 10]

- 5** In multicellular eukaryotes, gene regulation is important in development. Different sets of genes need to be expressed in different cells, at the right times, and in the right sequence for organisms to develop correctly.

In eukaryotes, a specific transcription factor or activator binds to an enhancer to regulate the expression of genes coding for specific proteins in different tissues.

- (a) (i)** Explain the meaning of the term 'enhancer'.

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.....[1]

- (ii)** Using the information given, suggest a reason why activators are referred to as 'specific' transcription factors.

.....
.....[1]

- (b) The *PITX1* gene codes for a protein that plays a role in the development of pelvic spines of fishes like the three-spine stickleback, *Gasterosteus aculeatus*. *PITX1* also plays critical roles in controlling the development of the fish body like the formation of the jaws, pelvis and pituitary glands.

Fig. 5.1 shows a diagram of the *PITX1* gene and the associated regions.

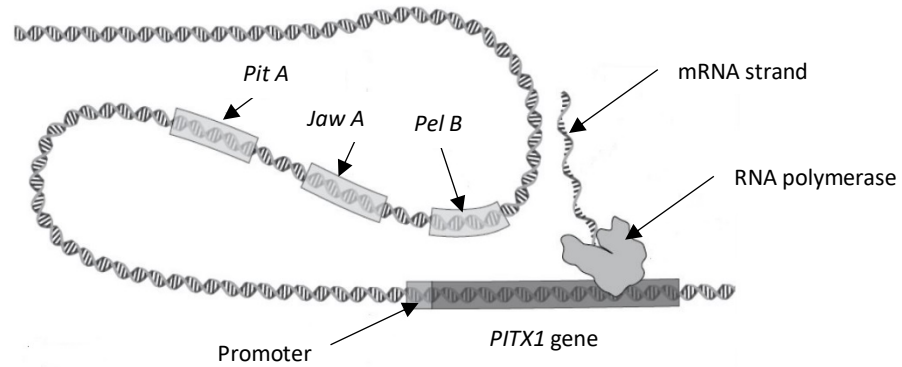


Fig. 5.1

- (i) The three regions, *Pel B*, *Jaw A*, *Pit A* are important for the expression of the *PITX1* gene.

Describe how the binding of an activator like *Pel B* at the *Pel B* region can affect gene expression.

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[2]

- (ii) Explain how the structure of the *Pel B* activator enables it to bind to the *Pel B* region.

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[1]

- (iii) Assuming that cells in the pelvis and eyes of the stickleback contain the same *PITX1* gene and the same three regions as shown in Fig. 5.1, explain why eyes of the stickleback do not have spines.

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[2]

- (c) Scientists researching on different species of sticklebacks living in the marine environments and in freshwater lakes found that they differed in the presence or absence of the pelvic spines as shown in Fig. 5.2.

Molecular genetic analysis found that these two groups of fishes exhibited very different levels of expression for the *PITX1* gene. Fig. 5.3 shows the level of expression (in arbitrary units or A.U.) of the *PITX1* gene in the pelvic regions for the fishes from both habitats.



Fig. 5.2

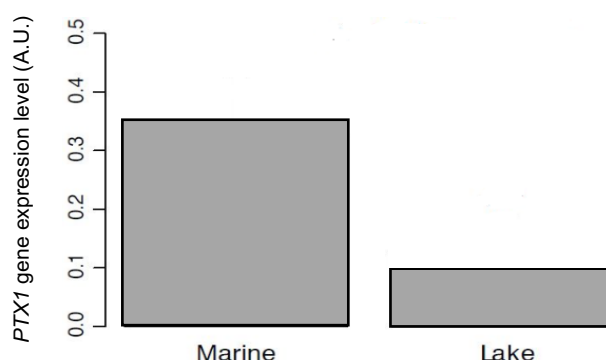


Fig. 5.3

- (i) With reference to Fig. 5.3, suggest a reason for the absence of pelvic spines in freshwater stickleback.

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.....[2]

- (ii) In freshwater habitats, dragonfly larva are much larger in size than young sticklebacks. They predate on the young sticklebacks by grabbing on to protruding parts of the fish.

Suggest how the absence of pelvic spines aids in the survival of the young sticklebacks.

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.....[1]

[Total: 10]

6 The patty pan squash plant, *Cucurbita pepo*, produces edible fruits that vary in colour.

(a) The colour of the fruits is controlled by two genes, **A/a** and **B/b**, that occur on different chromosomes.

- Allele **A** produces a white fruit colour.
- Allele **a** does not produce a colour by itself but allows the colours coded by gene **B/b** to show in the phenotype.
- Allele **B** produces a yellow fruit colour.
- Allele **b** produces a green fruit colour.

In a dihybrid cross, a pure-breeding white parent plant was crossed with another pure-breeding green parent plant. All the resulting F_1 plants produced white fruits.

The F_1 plants were then crossed with each other to obtain the F_2 generation.

(i) State the name for this type of gene interaction.

.....[1]

(ii) Draw a genetic diagram to show the cross of the F_1 plants to obtain the F_2 generation.

[4]

- (b) Explain how different genotypes give rise to a white phenotype.

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.....[3]

- (c) Explain how you would determine if the genotype of a white squash was homozygous dominant or homozygous recessive at gene locus **B**, assuming that it is heterozygous at gene locus **A**.

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.....[2]

[Total: 10]

- 7 (a) Chemiosmosis is the term used to describe the synthesis of ATP using a proton gradient across a membrane in a mitochondrion or chloroplast. It was first demonstrated by Peter Mitchell in 1961.

- In some of his experiments, Peter Mitchell used mitochondria that had been isolated from cells.
- The mitochondria were kept in liquid, in glass dishes, to which ADP, Pi and other substances were added.
- The outer mitochondrial membrane is freely permeable to proton entry.
- The temperature, pH and water potential were kept constant.
- After a period of time, he checked for the presence of ATP.

The contents of some of the dishes are shown in the table below.

Complete the table using a tick (✓) if ATP was produced, and a cross (✗) if no ATP was produced. [2]

dish contents	ATP produced
mitochondria + ADP + Pi + acetyl CoA + oxygen	
mitochondria + ADP + Pi + acetyl CoA	
mitochondria + ADP + Pi + low concentration of protons (H^+)	
mitochondria + ADP + Pi + high concentration of protons (H^+)	

- (b) Explain the consequences to a mitochondrion if the water potential of the liquid in the dishes is higher than the water potential of the mitochondrial matrix.

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.....[2]

(c) Fig. 7.1 is a transmission electron micrograph of a chloroplast.

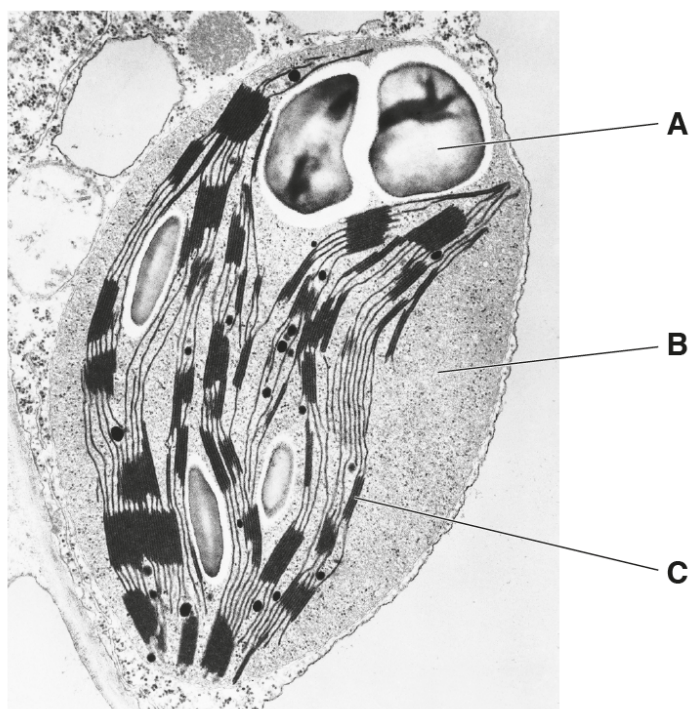


Fig. 7.1

Many compounds and structures involved in photosynthesis are located in a chloroplast.

Using the labels **A**, **B** or **C**, complete Table 7.1 to show the location of four of these compounds or structures.

You may use each of the letters **A**, **B** and **C** once, more than once, or not at all.

Table 7.1

compound or structure	location
ATP synthase	
rubisco	
starch grain	
phospholipid bilayer	

[2]

- (d) The light dependent stage of photosynthesis in a suspension of isolated chloroplasts can be investigated using the Hill reaction.

Dichlorophenolindophenol (DCPIP) can be used to follow the process. DCPIP is a blue dye which turns colourless when it is reduced by accepting hydrogen and electrons.

oxidised DCPIP $\longrightarrow$ reduced DCPIP

blue $\longrightarrow$ colourless

- (i) DCPIP is an artificial hydrogen acceptor that can be used in the Hill reaction.

Name the natural hydrogen acceptor found in chloroplasts that is replaced by DCPIP in the Hill reaction.

.....[1]

- (ii) Outline the way in which hydrogen is made available to reduce the hydrogen acceptor in the light dependent stage of photosynthesis.

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.....[2]

- (iii) Predict **and** explain the effect on the concentration of RuBP in the chloroplasts if DCPIP becomes reduced instead of the natural hydrogen acceptor.

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.....[1]

[Total: 10]

8 Sickle cell anaemia is an autosomal recessive inherited disorder.

- The Hb^A allele codes for the normal β -globin polypeptide of haemoglobin.
- The Hb^S allele, caused by a base substitution mutation, codes for an abnormal β -globin polypeptide.
- The base substitution results in the amino acid glutamate, which has a charged R group, to be replaced by valine, which has a non-polar R group, in the polypeptide.

The abnormal haemoglobin molecules (HbS) form fibres in low partial pressures of oxygen (pO_2). The fibres cause red blood cells to become sickle shaped and the cells can block blood capillaries.

Individuals with adult haemoglobin molecules that are all abnormal (HbS) have sickle cell anaemia. This is a painful chronic condition that can be life-threatening.

(a) Explain why this mutation causes the HbS to form fibres.

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.....[2]

(b) People who are heterozygous (Hb^A Hb^S) have sickle cell trait (SCT). For a child to inherit sickle cell anaemia (Hb^S Hb^S), both parents must have SCT. A genetic screening program is available for sickle cell anaemia and SCT.

- (i)** To test for the presence of Hb^S, DNA is extracted and the polymerase chain reaction (PCR) is carried out with two specific sets of primers. One set of primers (normal-specific primers) detects Hb^A allele while the other set of primers (mutant-specific primers) detects Hb^S allele.

Explain:

- why primers are used in PCR
- how the use of two specific sets of primers allows the amplification of the normal, sickle cell anaemia and SCT genotypes.

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..... [2]

- (ii) Gel electrophoresis is carried out on the products of the PCRs.

Fig. 8.1 includes the results for two individuals, **A** and **B**, tested for the sickle cell allele.

- Each lane has an 860 base pair (bp) band to indicate the test is valid.
- Lane 1 is a control lane with a 207bp band for an individual with known normal phenotype.
- Lane 2 is a control lane with a 207bp band for an individual with known sickle cell anaemia phenotype.
- Lanes 1, 3 and 5 contain DNA from the PCR that used normal-specific set of primers.
- Lanes 2, 4 and 6 contain DNA from the PCR that used mutant-specific set of primers.

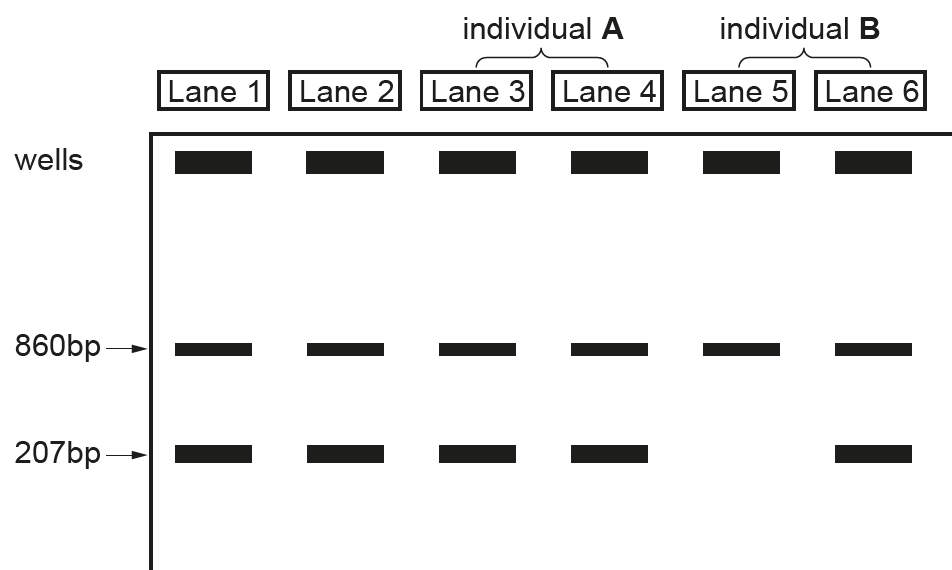


Fig. 8.1

Deduce the genotypes and phenotypes of individuals **A** and **B** in Fig. 4.1.

A:

.....

B:

.....[2]

(c) Genetically modified stem cell transplant is a common form of gene therapy used to treat sickle cell anaemia. It involves the following steps:

1. Removing haematopoietic stem cells (HSCs) from the patient
2. Genetically modifying the genome of the HSCs by inserting normal Hb^A allele
3. Reintroducing the modified stem cells back into the patient.

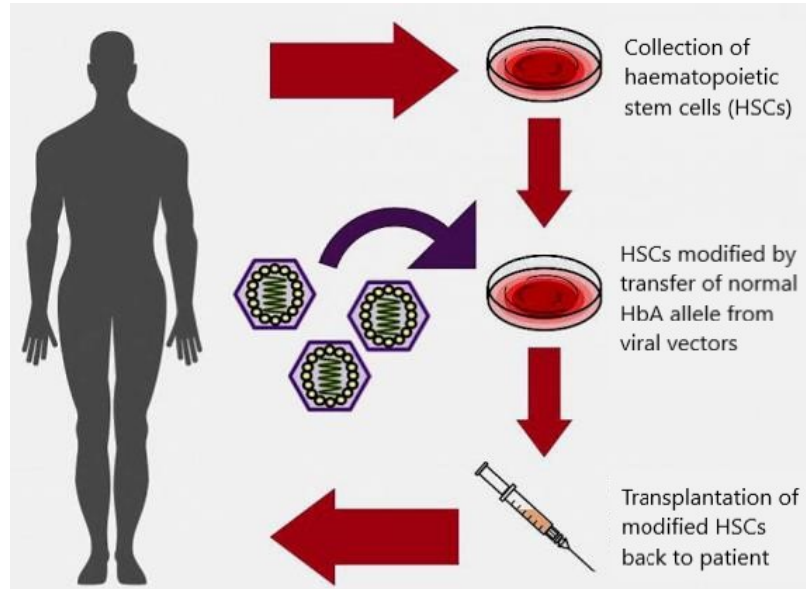


Fig. 8.2

(i) Describe the characteristics of haematopoietic stem cells that enable them to be used for stem cell therapy.

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.....[3]

(ii) Suggest an advantage of the specific type of stem cell therapy shown in Fig. 8.2.

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.....[1]

[Total: 10]

- 9 Different signalling pathways generate calcium (Ca^{2+}) signals which regulate many cellular functions, e.g. smooth muscle contraction.

Fig. 9.1 shows the inositol trisphosphate/calcium ($\text{IP}_3/\text{Ca}^{2+}$) signalling pathway. Phosphatidylinositol-bisphosphate (PIP_2) is hydrolysed into IP_3 and diacylglycerol (DAG) by phospholipase C.

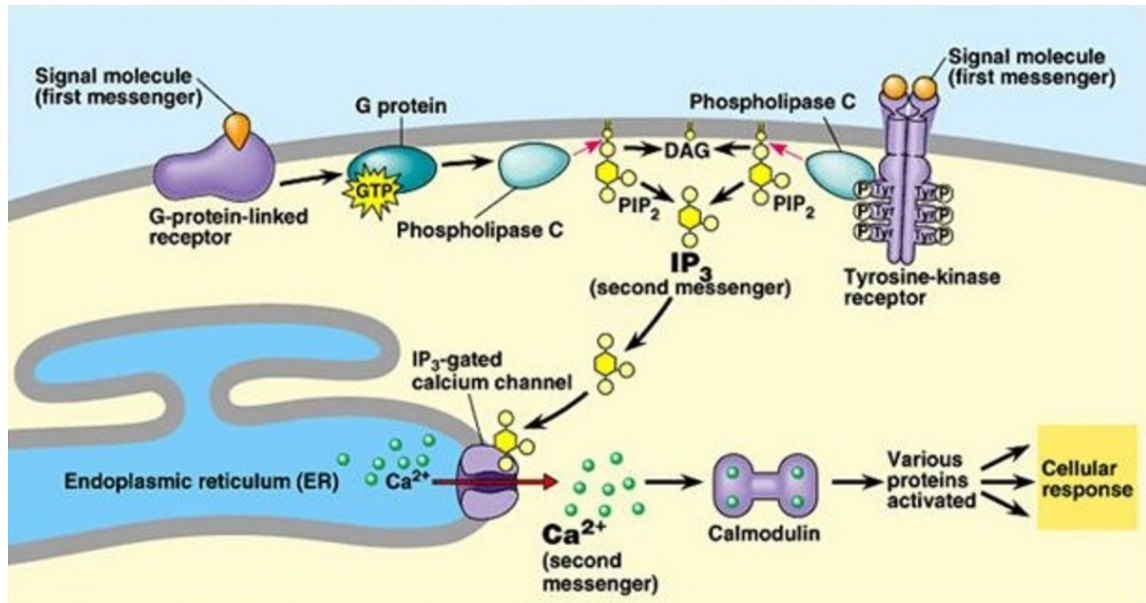


Fig. 9.1

(a) With reference to Fig. 9.1,

- (i) describe the structural differences of the two receptors.

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[2]

- (ii) define second messengers and explain their role in signal transduction.

define

.....

role

.....
[2]

- (iii) outline two stages where signal amplification may occur.

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.....[2]

- (iv) Dysregulation of the $\text{IP}_3/\text{Ca}^{2+}$ signalling pathway can lead to many different possible human diseases. Hypertension is caused by increased smooth muscle contraction due to enhanced $\text{IP}_3/\text{Ca}^{2+}$ signalling.

Suggest the type of mutation in phospholipase C that could lead to hypertension.

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.....[2]

- (b) Calcium binding by calmodulin exhibits considerable cooperativity, making calmodulin an unusual example of a monomeric (single-chain) cooperative-binding protein with four calcium-binding sites.

Fig. 9.2 shows calmodulin without calcium (left), and calmodulin with calcium (right). Sites that bind target proteins are indicated by the stars (*).

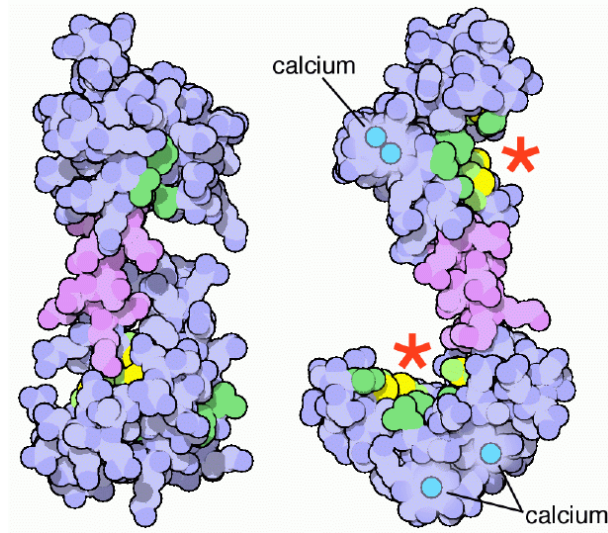


Fig. 9.2

With reference to Fig. 9.2, suggest how cooperative binding of Ca^{2+} ions to calmodulin is necessary for calmodulin activity.

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[Total: 10]

- 10 (a) Fig. 10.1 is a diagram that shows the structure of an antibody molecule.

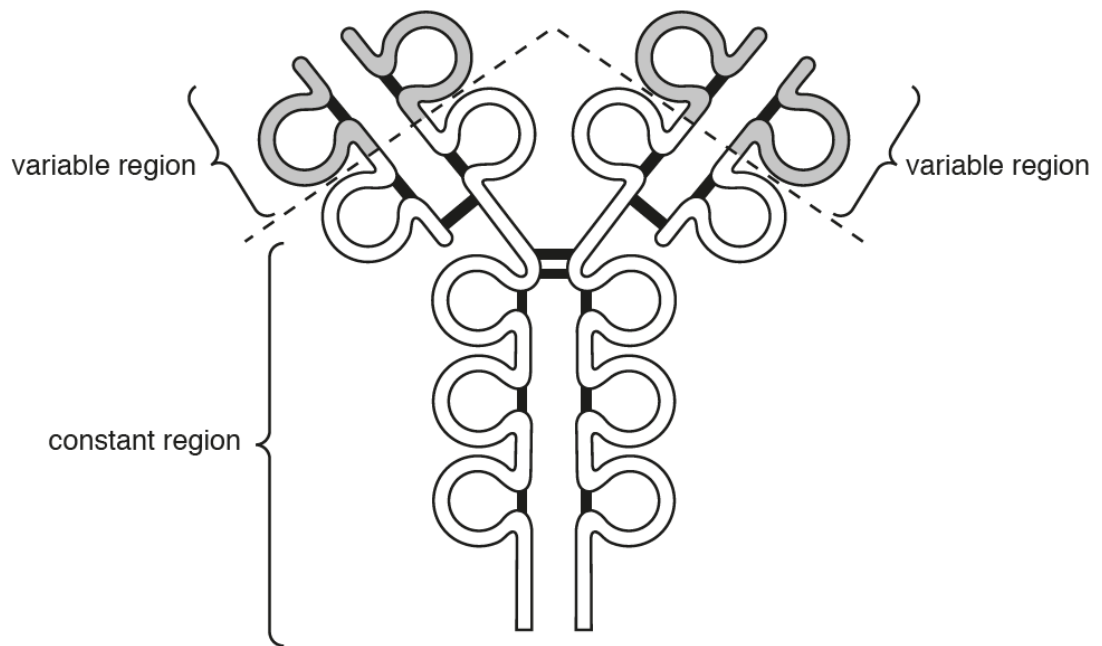


Fig. 10.1

Use Fig. 10.1 to explain how the structure of the variable region of an antibody molecule is related to its function.

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.....[2]

- (b) Fig. 10.2 shows a process that occurs to heavy chain gene on human chromosome 14.

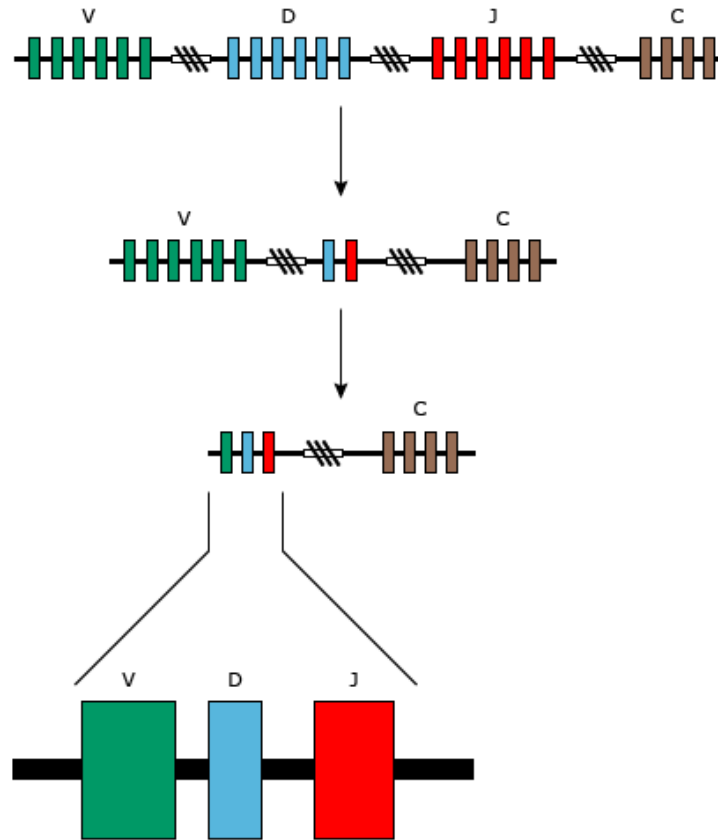


Fig. 10.2

- (i) Outline the process shown in Fig. 10.2.

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- (ii) Explain the significance of the above process.

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.....[1]

[Total: 5]

- 11 Fig. 11.1 shows the relationship between the density of coral skeletons and the rate of precipitation of key minerals for the coral skeleton. The different shapes represent coral samples obtained from different locations in the world.

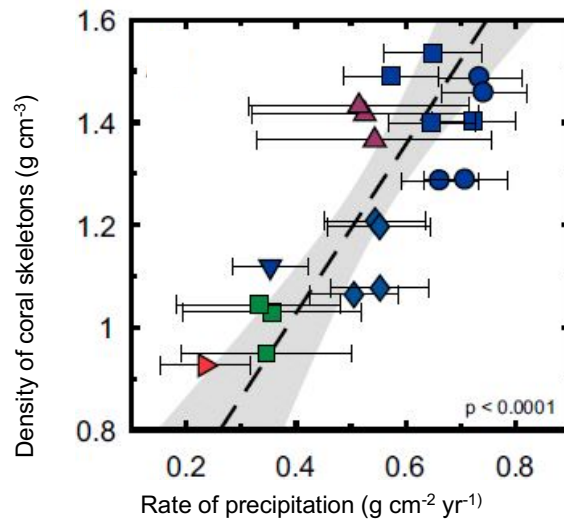


Fig. 11.1

- (a) (i) Describe the general relationship between density of coral skeletons and rate of precipitation of key minerals for the coral skeleton.

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[1]

- (ii) An increase in carbon dioxide concentration in the atmosphere has resulted in higher concentrations of carbon dioxide in the ocean. This has caused a decrease in the pH of the ocean and has resulted in ocean acidification.

Explain how ocean acidification has affected corals.

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[1]

- (b) Scientists are studying seaweeds such as *Laminaria hyperborea* because they absorb a large quantity of carbon dioxide during photosynthesis. This may help to increase the pH of the ocean and reverse ocean acidification.

In the laboratory, *Laminaria hyperborea* was grown in water with different pH values. All other variables, including temperature and light, were standardised.

The mean rate of photosynthesis was calculated over a 24 hour period for each pH value.

The results are shown in Fig. 11.2.

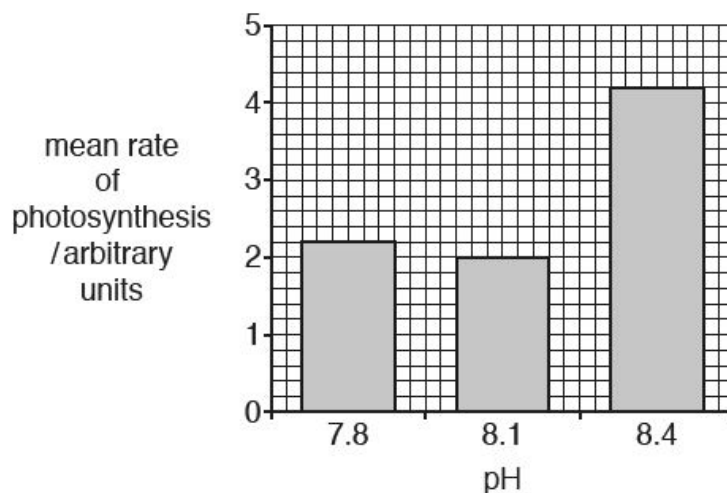


Fig. 11.2

The lower pH values on Fig. 11.2 represent ocean acidification.

With reference to Fig. 11.2, suggest why the results for the lower pH values do **not** fully support the idea that seaweeds can help to reduce ocean acidification.

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[Total: 5]

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